

IDENTIFICATIONCODE : GI-5-S1-EC-ADM
ECTS : 2**HOURS**Cours : 0h
TD : 6h
TP : 12h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 20h
Travail personnel : 20h
Total : 40h**ASSESSMENT METHOD**

continuous assessment

TEACHING AIDS

Lecture book and exercises

TEACHING LANGUAGE

French

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The Multicriteria Decision Aid course covers methods and tools that help a decision-maker make a complex decision involving multiple criteria, often contradictory or in conflict.

Three approaches will be covered in this course:

- The AHP method (Analytic Hierarchy Process);
- The TOPSIS method (Technique for Order Preference by Similarity to Ideal Solution);
- Multicriteria optimization.

CONTENTTD1 2h introduction ADM + AHP
TP1 4h TP AHP CC1
TD2 2h OPTIM + TOPSIS
TP2 4h TP TOPSIS CC2
TD3 2h TD Lexico + Pareto
TP3 4h TP Pareto CC3**BIBLIOGRAPHY**Bouyssou D. et Roy B. : Aide multicritère à la décision : méthode et cas, Economica, 1993.
Carlier J. et Chrétienne P. : Problèmes d'ordonnancement : modélisation / complexité / algorithmes, Masson, 1988.
Garfinkel R.S. et Nemhauser : Integer programming, Wiley Interscience, 1972.
Roy B. : Méthodologie multicritère d'aide à la décision, Economica, 1985.**PRE-REQUISITES**

Monocriterion decision support (Linear Programming), basic and advanced production management